ERA 5 calibration to Elexon power

S. Gomez[1]

[1]School of Mathematics, University of Edinburgh

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Calibration of long term series of wind power

Calibration of long term series of wind power

- Using 6 years of hourly data (2019-2024) to fit models.
- Model are fitted and WF level and then aggregated to GB level.
- Model validation with year 2025 data.
- Also accuracy during low wind events is assessed.

Next

- Reconstruct long term series of wind power
- Options
  - Iain: Fix fleet and use ERA5 since 80s
  - Hannah: Use climate projections (daily series downscaled to hourly)
  - From REPD we can get upcoming wind farms (Accepted and Awaiting Construction)

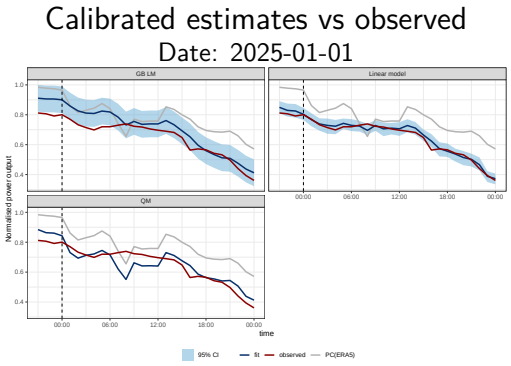
Models fitted

- Quantile mapping
- Bayesian model with smooth effects (terrain + temporal + turbine type + wind speed)
- Aggregated GB onshore/offshore models for benchmark

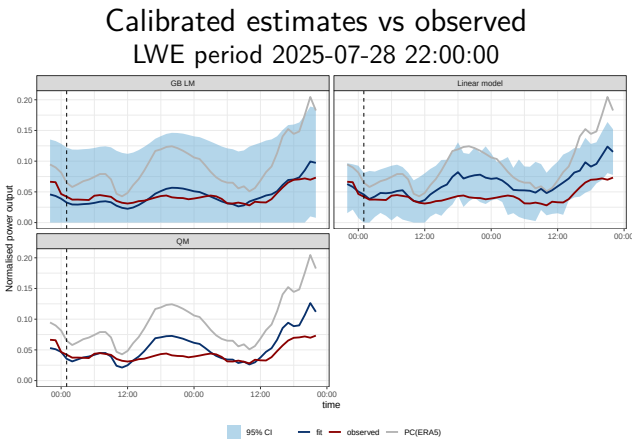
model	RMSE	MAE	Bias
Generic PC	0.231	0.162	0.114
QM	0.174	0.117	0.000
GB LM	0.168	0.118	0.016
Linear model	0.149	0.107	0.002

Model vs observed in 2025

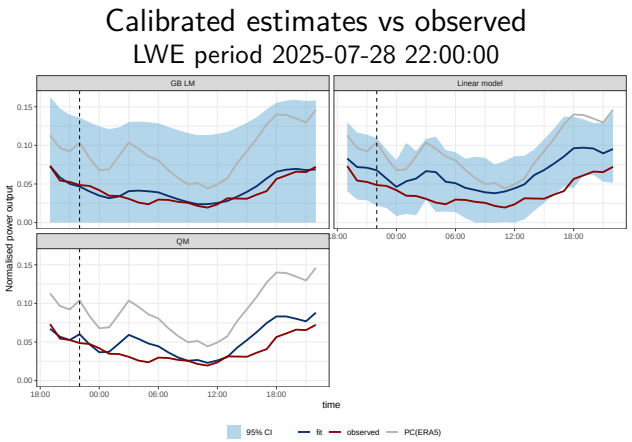
- Coverage remains similar to ST exercise
- GB model looks better on aggregated series
- QM tends to have better accuracy in low regime than LM



Model vs observed in 2025



Model vs observed in 2025



Performance at aggregate GB level

model	RMSE	MAE	MAPE	Bias
Generic PC	0.138	0.124	40.902	0.124
QM	0.036	0.026	8.905	0.007
GB LM	0.033	0.025	9.310	0.000
Linear model	0.029	0.023	9.544	-0.004

Performance by regime

p_group3	model	RMSE	MAE	MAPE	Bias
high	Generic PC	0.257	0.193	148.384	0.142
high	QM	0.199	0.145	82.048	0.004
high	GB LM	0.189	0.141	134.597	0.004
high	Linear model	0.168	0.127	115.051	-0.013
mid	Generic PC	0.233	0.164	425.756	0.116
mid	QM	0.170	0.114	207.731	0.001
mid	GB LM	0.166	0.117	396.410	0.025
mid	Linear model	0.147	0.105	398.135	0.008
low	Generic PC	0.158	0.096	941.417	0.052
low	QM	0.121	0.068	444.550	-0.009
low	GB LM	0.120	0.074	797.112	0.015
low	Linear model	0.108	0.070	931.196	0.015

Performance during low wind events

lwe	model	RMSE	MAE	MAPE	Bias
FALSE	Generic PC	0.234	0.166	376.057	0.117
FALSE	QM	0.176	0.120	187.725	0.000
FALSE	GB LM	0.170	0.120	329.790	0.016
FALSE	Linear model	0.151	0.109	341.442	0.001
TRUE	Generic PC	0.086	0.046	1430.679	0.027
TRUE	Linear model	0.067	0.043	1721.594	0.021
TRUE	GB LM	0.064	0.040	1604.130	0.017
TRUE	QM	0.059	0.030	577.859	-0.002

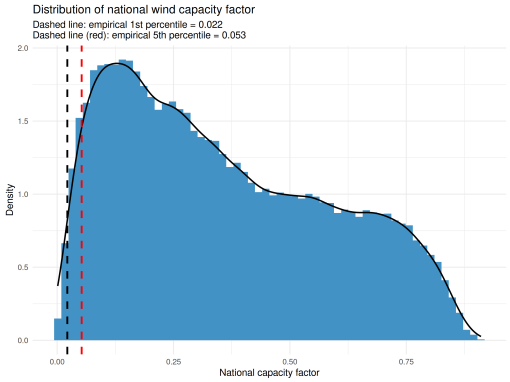
Low wind events EDA

Low wind events EDA

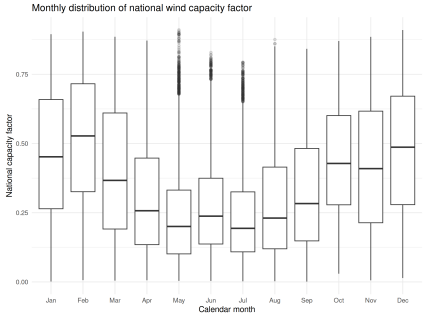
- Important at aggregate GB level for system operation and security of supply.
- We define low wind events as periods of time when the total GB wind power generation is below a certain threshold (e.g., 5% of installed capacity).
- We can analyze the distribution of these events in terms of their duration, frequency, and severity (e.g., cumulative shortfall).
- We can also compare the observed low wind events with those simulated by the ERA5 reanalysis data to assess the accuracy of the reanalysis in capturing these events.
- Ideally we'll model shortfall events, considering demand and other generation sources, but for now we focus on wind generation only.

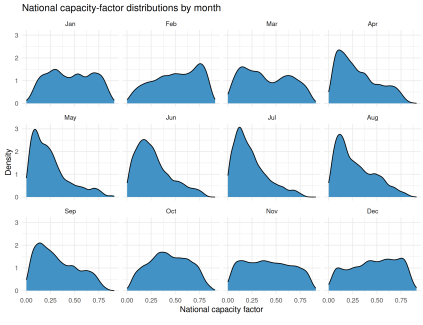
Capacity Factor distribution

- Historical distribution of national capacity factor (CF) for wind power in GB.
- The CF is defined as the ratio of actual power output to the maximum possible output in one hour
- Data spans from 2019 up to 2025 for 147 locations across GB.

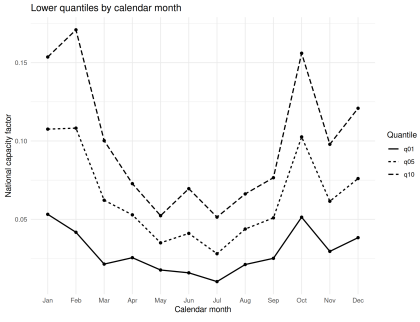


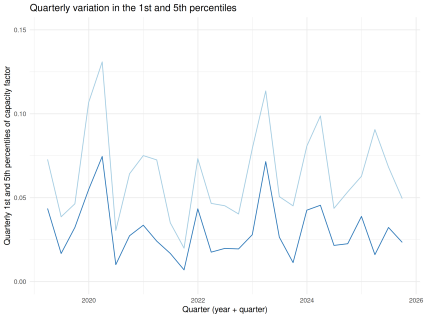
Seasonal variation of CF



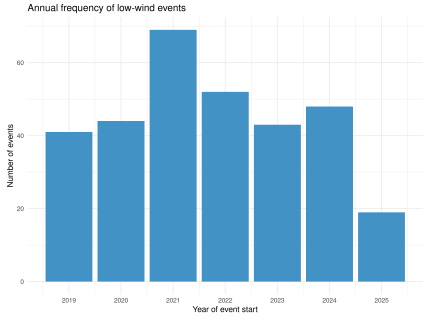


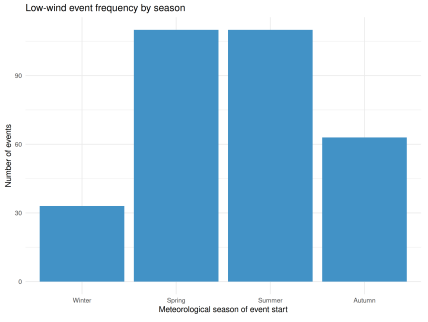
CF quantiles seasonality



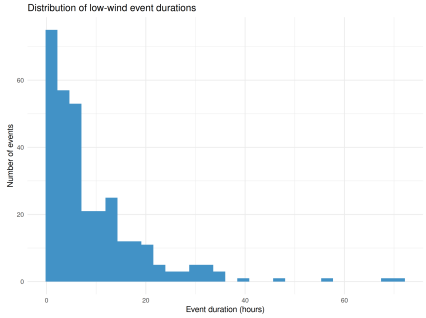


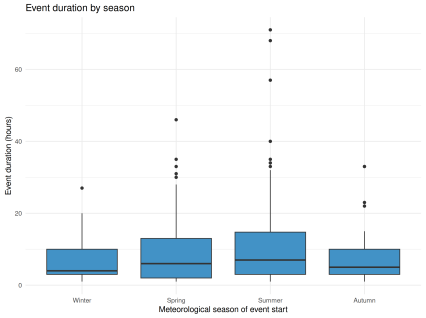
Frequency of events below 5% CF



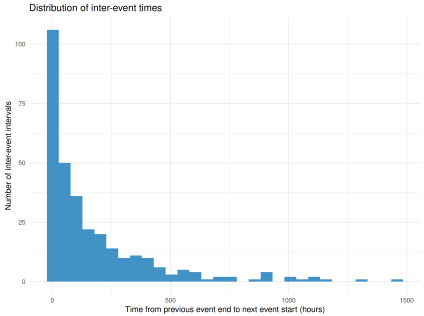


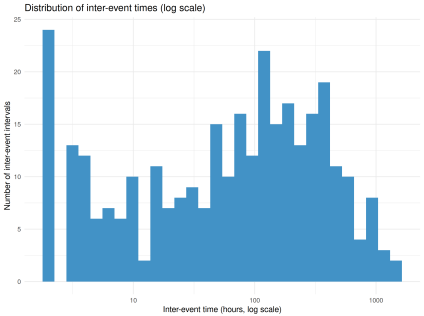
Duration of events below 5% CF



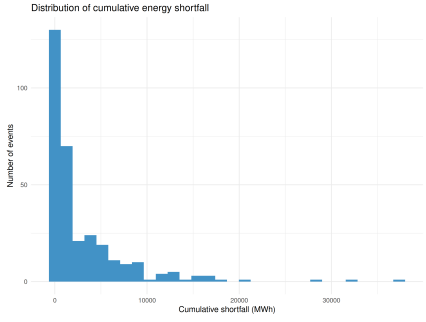


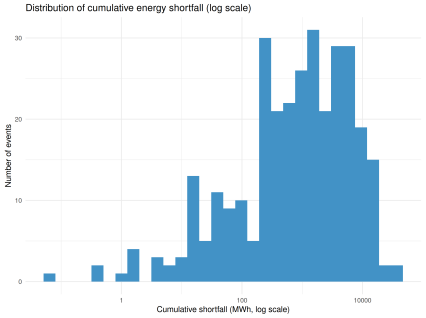
Interevent time of events below 5% CF



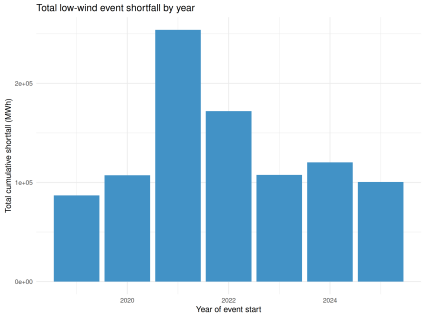


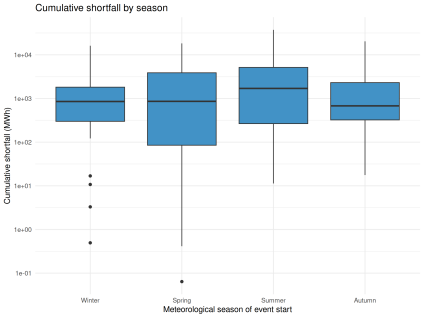
Severity of events below 5% CF





Severity of events below 5% CF, annual and seasonal variation





Worst low wind events

